

ORTONVILLE, MN, USA

WMO: 727515

Lat: 45.306N

Lon: 96.424W

Elev: 335

StdP: 97.36

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04982

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-22.5	-28.7	0.3	-23.8	-27.0	0.3	-22.4	13.4	-9.3	12.1	-9.4	3.4	330	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.5	22.6	31.1	22.1	29.0	21.0	24.6	30.1	23.6	28.7	22.5	27.6	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.2	28.1	22.1	17.5	27.4	21.0	16.4	26.4	76.2	30.4	72.2	28.3	68.2	27.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.8	10.5	9.0		-28.4	35.8	3.2	1.9	-30.7	37.1	-32.6	38.2	-34.4	39.3	-36.7	40.7
			DB	-28.7	26.3	3.1	1.3	-30.9	27.3	-32.8	28.0	-34.5	28.8	-36.8	29.7
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.7	-8.6	-0.7	7.3	14.5	20.1	22.8	21.1	16.9	8.9	1.1	-7.1	(6)
(7)		DBStd	12.82	7.46	7.16	7.27	5.89	4.89	3.54	3.11	3.09	4.50	5.27	6.29	6.87	(7)
(8)		HDD10.0	2489	611	520	337	117	16	0	0	0	3	85	271	529	(8)
(9)		HDD18.3	4397	869	753	590	332	139	22	4	9	78	297	516	788	(9)
(10)		CDD10.0	1508	0	0	6	37	154	304	396	346	211	50	6	0	(10)
(11)		CDD18.3	374	0	0	0	1	19	76	142	96	37	3	0	0	(11)
(12)		CDH23.3	3463	0	0	2	33	231	678	1303	799	364	52	1	0	(12)
(13)		CDH26.7	1141	0	0	0	8	75	209	467	249	117	15	0	0	(13)
(14)	Wind	WSAvg	4.0	4.1	4.3	4.5	5.0	4.6	3.8	3.1	3.1	3.6	4.1	4.0	4.1	(14)
(15)	Precipitation	PrecAvg	634	17	18	33	64	71	98	98	68	71	59	25	16	(15)
(16)		PrecMax	819	43	40	93	172	130	191	200	134	172	173	97	46	(16)
(17)		PrecMin	404	0	3	6	8	22	34	41	14	7	6	0	2	(17)
(18)		PrecStd	109	12	11	21	36	26	44	42	35	39	45	25	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	11.2	22.0	27.5	32.4	32.9	35.2	33.9	32.6	28.8	20.3	10.0	(19)
(20)			MCWB	3.8	7.5	13.1	15.2	19.0	22.2	24.6	23.4	21.5	17.1	12.2	4.8	(20)
(21)		2%	DB	4.1	7.1	17.0	22.7	28.0	31.2	32.5	31.3	30.0	23.7	16.4	6.3	(21)
(22)			MCWB	1.7	3.9	11.0	12.5	17.4	21.3	23.7	23.1	20.1	14.9	9.8	2.7	(22)
(23)		5%	DB	2.5	3.8	12.5	20.0	26.0	28.9	31.2	29.0	27.5	20.1	12.7	3.7	(23)
(24)			MCWB	0.7	1.9	8.4	11.5	16.2	20.3	23.2	21.7	18.9	13.1	7.9	1.4	(24)
(25)		10%	DB	1.1	2.2	8.9	17.3	22.8	27.4	29.0	27.6	25.1	17.4	9.8	2.1	(25)
(26)			MCWB	-0.1	0.8	6.1	10.2	14.7	19.2	21.9	20.7	18.0	11.5	6.3	0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	7.8	14.8	16.3	21.4	24.5	26.6	25.4	23.6	19.0	13.7	5.5	(27)
(28)			MCDB	7.7	10.8	18.2	24.3	29.3	30.5	31.9	31.3	29.0	25.9	18.7	9.0	(28)
(29)		2%	WB	2.1	4.0	11.5	14.1	19.1	22.9	25.2	24.1	21.9	16.0	10.7	3.2	(29)
(30)			MCDB	4.2	6.4	16.1	20.8	25.6	28.6	30.8	29.3	27.6	22.0	15.4	5.8	(30)
(31)		5%	WB	0.8	2.2	8.5	12.4	17.6	21.7	24.1	23.0	20.5	13.9	8.2	1.7	(31)
(32)			MCDB	2.3	4.0	12.2	18.4	23.6	27.3	29.2	27.6	25.5	19.2	12.5	3.6	(32)
(33)		10%	WB	-0.3	0.4	5.9	10.7	16.1	20.3	23.0	21.8	19.1	12.1	6.1	0.3	(33)
(34)			MCDB	0.8	2.0	9.0	16.2	21.5	25.5	27.9	26.3	23.6	16.3	9.7	1.7	(34)
(35)	Mean Daily Temperature Range	MDBR	9.2	9.5	9.4	10.8	11.1	10.4	10.8	10.8	11.7	10.4	9.6	8.2	(35)	
(36)		5% DB	MCDBR	11.2	11.3	14.6	16.6	15.4	13.4	12.7	12.9	14.9	15.8	14.9	11.6	(36)
(37)			MCWBR	9.1	8.8	9.4	8.5	7.2	6.2	6.1	6.3	6.8	8.4	9.3	8.8	(37)
(38)		5% WB	MCDBR	10.3	10.8	14.2	14.2	12.9	11.1	11.3	11.5	12.5	13.5	14.1	10.3	(38)
(39)			MCWBR	8.5	8.4	9.4	8.1	7.2	6.3	6.3	6.4	7.0	7.9	9.2	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.267	0.277	0.308	0.351	0.371	0.385	0.389	0.395	0.359	0.320	0.287	0.265	(40)	
(41)		taud	2.429	2.417	2.440	2.359	2.374	2.389	2.365	2.331	2.417	2.502	2.507	2.447	(41)	
(42)		Ebn at Noon	859	916	929	908	897	883	874	854	860	854	829	823	(42)	
(43)		Edh at Noon	70	86	97	115	118	117	118	118	99	79	65	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.77	2.85	3.80	4.82	5.52	6.20	6.57	5.48	4.23	2.71	1.77	1.30	(44)	
(45)		RadStd	0.19	0.31	0.35	0.38	0.39	0.37	0.25	0.46	0.30	0.35	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page